

Risk Calculation System: Comprehensive Test Plan

1. Introduction and Objectives

This test plan outlines the rigorous validation framework for the Risk Calculation System. The objective is to verify the accuracy, robustness, and theoretical soundness of the implemented models for computing Value at Risk (VaR) and Expected Shortfall (ES). In accordance with sound model risk management practices, this plan not only tests the core functional requirements (Monte Carlo, Historical, and Parametric methods) but also integrates Risk Identification, Stress Testing, and Model Risk Management protocols to expose and document model limitations.

2. Input and Data Parsing Validation

Ensure the system robustly handles arbitrary sets of securities and their respective market data.

- **Test 2.1: Portfolio Ingestion**
 - *Action:* Input a standardized file containing a mixed portfolio of stocks and options.
 - *Expected Result:* System accurately parses positions, identifies instrument types, and extracts necessary option parameters (strike, maturity, option type).
- **Test 2.2: Historical Data Alignment**
 - *Action:* Supply historical price data with missing dates or non-synchronous trading days.
 - *Expected Result:* System correctly aligns time series, implements forward-fill or appropriate interpolation, and flags significant data gaps.

3. Risk Identification and Core Calibrations

Verify underlying statistical assumptions and intermediate calibration steps before risk calculation.

- **Test 3.1: Covariance Matrix PSD Check**
 - *Action:* Calculate the covariance matrix Σ for a large-dimensional stock portfolio.
 - *Expected Result:* The resulting matrix is symmetric and positive semi-definite (PSD).
- **Test 3.2: Distributional Assumption Testing (Risk Identification)**
 - *Action:* Perform a statistical test (e.g., Jarque-Bera) on the historical log-returns.
 - *Expected Result:* System outputs p-values identifying deviations from normality (fat tails/skewness), documenting the theoretical limitations of the Parametric VaR approach.
- **Test 3.3: Option Volatility Calibration**
 - *Action:* Input both historical asset volatility and implied volatility for an option position.
 - *Expected Result:* The system strictly utilizes implied volatility to price the option, while appropriately modeling volatility dynamics for risk simulations.

4. Risk Measurement and Assessment

Validate the quantitative accuracy of the primary risk metrics.

- **Test 4.1: Parametric VaR Accuracy**

- *Action:* Compute VaR at confidence level α using the analytical formula:

$$\text{VaR}_\alpha = -(\mu + z_{1-\alpha}\sigma) \times V$$

- *Expected Result:* Output exactly matches manually calculated benchmarks using the validated covariance matrix.

- **Test 4.2: Historical VaR and ES Execution**

- *Action:* Execute historical simulation. Verify that ES is computed strictly as the conditional expectation of losses exceeding VaR:

$$\text{ES}_\alpha = E[L | L > \text{VaR}_\alpha]$$

- *Expected Result:* VaR matches the exact empirical quantile; ES accurately averages the tail losses.

- **Test 4.3: Monte Carlo Convergence**

- *Action:* Run the Monte Carlo engine for a non-linear portfolio using $N = 1,000, 10,000$, and $100,000$ paths.
- *Expected Result:* The standard error of the VaR estimate decreases at a rate proportional to $1/\sqrt{N}$, confirming algorithmic convergence.

5. Model Backtesting Validation

Assess the predictive power of the VaR models using historical out-of-sample data.

- **Test 5.1: Exception Tracking**

- *Action:* Run a rolling-window 99% VaR over a 252-day historical period and compare against actual next-day P&L.
- *Expected Result:* The system correctly flags instances where actual loss exceeds the predicted VaR.

- **Test 5.2: Statistical Validity of Exceptions**

- *Action:* Evaluate the exception rate against the designated confidence level using an unconditional coverage test.
- *Expected Result:* The exception rate statistically aligns with the $1 - \alpha$ expectation; significant deviations trigger a required justification in the Model Documentation.

6. Scenario Analysis and Stress Testing

Evaluate system performance under extreme market conditions to identify vulnerabilities not captured by standard VaR.

- **Test 6.1: Historical Stress Scenarios**

- *Action:* Inject historical data from periods of severe market dislocation (e.g., the 2008 financial crisis).
- *Expected Result:* Historical ES reflects severe tail risk, and models run without computational failure despite extreme volatility spikes.

- **Test 6.2: Hypothetical Shock (Scenario Analysis)**

- *Action:* Apply a deterministic, instantaneous shock to the portfolio (e.g., a 20% drop in equities coupled with a 50% relative increase in implied volatility).

- *Expected Result:* The system isolates and re-prices the portfolio, returning the exact magnitude of the stress loss to inform capital adequacy bounds.

7. Model Risk Management (MRM) Benchmarking

Ensure the implementation aligns with industry-standard control frameworks.

- **Test 7.1: Greeks-Based Benchmarking**

- *Action:* Compare the full-revaluation Monte Carlo VaR of an options portfolio against a first-order Delta-Normal parametric approximation.
- *Expected Result:* Results are within an acceptable tolerance for linear portfolios, while divergence is clearly reported for highly non-linear (gamma-heavy) portfolios, validating the necessity of the Monte Carlo approach.

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